

- **Title:** Enabling the Study of Lyric-Music Correspondence in Song: Encoding Linguistic Annotations in Humdrum Scores
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Lyrics are a ubiquitous and essential part of most of the world’s most popular forms of music. Unfortunately, computational musicology—like many areas of music research—has often given lyrics short shrift. Though music encoding standards (musicXML, humdrum, MEI, etc.) allow for the encoding of lyrics in a mode similar to conventional Western scores, many major collections of digital scores simply don’t include lyric encodings (e.g., Essen). Even when lyrics are included in datasets, there is generally little consistency or rigor: For example, no standards for capitalization, punctuation, dialectical pronunciations, or marking syntactic boundaries have been adopted in music research. Though computational musicology is a close cousin to computational linguistics, we have generally failed to consider how linguistic standards or analyses (for example, part-of-speech or syntactic dependency tags) might be applied to lyric data.

In this paper I present systematic standards and best practices for the *consistent* and detailed encoding of lyrics in humdrum-syntax data, including approaches to annotating “lyrics-specific” annotations like pronunciation, rhyme, parts-of-speech, and syntactic dependencies. I also demonstrate practical workflows for adding such annotations to existing datasets. I demonstrate these standards applied to humdrum-encoded datasets of popular music (MCFlow and CoCoPops) and classical music (Bach and Telemann cantatas).

Using my new lyric data, I analyze the structural relationships between lyrics and music. I investigate how musical grouping structures (phrases, subphrases, motives) can align—or conflict—with linguistic structures. In this talk, I will focus on two analyses: (1) how multi-syllable word boundaries are set to music, and particularly how classical music and vernacular music approach this differently, and (2) how syntactic boundaries in lyrics can conflict with musical boundaries to create enjambment-like effects. I document and classify a wide variety of such effects in both rock and hip hop music; For example, I show how rock singers and rappers use conjunctions (“and”, “but”, etc.) at the end of musical phrases to create tension and drive attention into the next phrase.